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The metrics of visual and haptic space based on parallelity judgements

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Abstract

It has long been known that the visually perceived positions of objects, in short visual space, are distorted with respect to the physical positions. On the basis of the observation that equidistance-alleys lie outside parallel-alleys, Luneburg (Mathematical Analysis of Binocular Vision, Princeton University Press, Princeton, NJ, 1947) proposed that visual space is a Riemannian space of constant negative curvature. Luneburg used this observation, along with some additional assumptions, as an axiom and deduced the metric of visual space theoretically. Many researchers have tried to verify Luneburg's model experimentally, but the results are ambiguous. Moreover, many assumptions that Luneburg made were proved wrong in the literature. In this paper we will derive metric models for both visual and haptic space based directly on visual and haptic experiments involving a parallelity task. From the metric we make predictions for the egocentric distance, frontoparallel horopters, parallel- and equidistance-alleys and compare them to the literature. We also compare the metric structures of haptic and visual space. Both models provide a good description of the data from the parallelity tasks, even though the so-called oblique effect observed experimentally has not yet been incorporated.
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1. Introduction

When we watch moving objects, normally their shape does not appear to change. Our general impression is that the shape and size of objects remain constant: for example, we do not see people grow or shrink as they walk towards us or away from us. The notion that vision provides us with a stable percept of the environment has led many researchers to believe that there exists also some stable internal representation often denoted visual space. However, visual estimates of equidistance, collinearity and parallelity have been shown to deviate considerably from physically correct. For instance, it was shown that when two rows of faint luminous lights extending depthwise appear to lie along two parallel lines (parallel-alleys), the actual positions are located along physically curved lines (Blumenfeld, 1913; Hillebrand, 1902). Furthermore, Blumenfeld (1913) found that these apparently parallel rows of lights do not appear to be equidistant; to be perceived as equidistant they must be arranged on differently curved lines, the so-called distance-alleys. Luneburg (1947) noted that in curved

geometries parallel lines are no longer equidistant and that shape constancy is preserved only in Riemannian spaces of constant curvature. Together with the observation that distance-alleys lie outside the parallel-alleys, this led Luneburg to propose that visual space is a Riemannian space of constant negative curvature. Many investigators have tried to establish the curvature based on this model (see Indow (1991) for an overview of this literature), but the results have been ambiguous. Moreover, direct tests of the constant curvature assumption revealed that this assumption generally does not hold (e.g. Foley, 1963, 1972; Higashiyama, 1984; Koenderink, Van Doorn, & Lappin, 2000). Experiments conducted out of doors showed that, in these circumstances, visual space is described more accurately with a Euclidean geometry and a simple distance relation (Battro, Pierro Netto, & Rozestraten, 1976; Battro, Reggini, & Karts, 1978; Wagner, 1985). So the following question remains: what is the geometry of visual space and is it metric?

In a previous study (Cuijpers, Kappers, & Koenderink, 2000b) we examined the structure of visual space with a parallelity task. In that experiment subjects were asked to adjust the orientation of a test bar until it appeared parallel to a reference bar. The bars were

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placed horizontally at eye height and rotated about the vertical axis through their centres. The locations of the bars and the orientation of the reference bar were varied systematically. The deviations from parallel were found to increase linearly with the separation angle between the bars. The deviations ranged from 10° to 40° for the maximum separation angle of 60° depending on the subject and were independent of the distance of the bars from the subject. Note that, despite these large errors, subjects were unaware of them and were convinced that the bars had approximately the same physical orientation in space. The size of the deviations was independent of the orientation of the reference bar except when the reference bar was aligned with or perpendicular to the orientation of the main axis of the experimental room; for these orientations the deviations were much smaller or even zero depending on the subject. Cuijpers, Kappers, and Koenderink (2001) showed that this ‘oblique effect’ was caused by external references. They found evidence that without external references the deviations would not vanish and would be completely independent of the orientation of the reference bar.

Very similar results were obtained for the haptic modality by Kappers (1999) and Kappers and Koenderink (1999). In these studies subjects were blindfolded and they adjusted the orientation of a test bar manually until it *felt* parallel to a reference bar. Both bars were located in a horizontal plane at waist height. The results are surprisingly similar: the deviations increase linearly with the lateral distance between the bars. The deviations ranged from 8° to 90.5° for the maximum lateral distance of 120 cm depending on the subject (see Kappers, 2002). Note that subjects were unaware of these enormous errors and felt disbelief when their settings were shown to them afterwards. A quick attempt to improve their performance revealed the same errors, however. For non-oblique orientations of the reference the deviations are smaller for most subjects; for subjects with very large deviations a reversed oblique effect was observed (Kappers, 2002). A change of the distance in depth of the bars from the subject had no effect on the deviations.

The parallelity tasks were designed bearing in mind that the haptic and visual perceptual spaces may not be metrical spaces. Parallelism is a non-metrical property that one can define without the need of a concept of length. This yields the so-called *affine* spaces. It appears that the structure of visual space is affine (Todd, Oomes, Koenderink, & Kappers, 2001). In differential geometry, the approach is rather different because all the structure is defined locally and needs to be integrated to obtain the global structure. As a consequence defining parallelism locally does not imply the existence of globally parallel lines. The simplest example is the surface of a sphere: all meridians intersect the equator at right angles but they intersect each other at the poles, so they are

locally parallel at the equator but not globally parallel. Local parallelism is defined by means of the so-called *affine connection* (the word *affine* here has no relation to the aforementioned term). The main goal of this paper is to demonstrate how the results from the parallelity tasks can be used to establish the affine connection of the corresponding perceptual spaces.

The use of differential geometric constructs is not new in the literature. To begin with, Luneburg (1947) postulated that visual space is a Riemannian space of constant curvature. Also, the much weaker assumption of affinely connected spaces has been proposed before. For example, Shimbo, Yamanoi, and Kawaguchi (1978) and Yamazaki (1987) showed how the affine connection can be determined from the classical alley experiments and developed a (non-Riemannian) model for visual space. Levin (2000) extended the application of affinely connected spaces to any continuous stimulus space. In this study, the affine connection is determined experimentally from perceptually equivalent transformations at various stimulus levels: for example, an equivalent rise in pitch at different levels of loudness and pitch. These transformations are needed to obtain estimates of parallel ‘vectors’. However, in our parallelity tasks, the parallel vectors are obtained directly by identifying the orientation of the bars with vectors of unknown length. This is a novel method for obtaining the affine connection and we will explain it in detail.

There are several other mathematical models that, in principle, could be applied to our data. Recently, Dzhaferov and Colonius (1999) proposed a technique to establish Finsler metrics for continuous stimulus spaces. This technique requires psychometric functions describing the discrimination probabilities among stimuli. Although it is an elegant extension to Fechner’s original empirical theory, it presupposes the existence of a metric which, in our case, is undesirable. Furthermore, we would need a lot more data to determine the required psychometric functions. Indeed, the experiments would become rather cumbersome, if we did try to measure them. Another approach is the use of Lie transformation groups (Hoffman, 1966; Dodwell, 1983). The affine connection defines a Lie transformation group so these methods are related, but the approach is rather different and we will not pursue it any further.

If the affine connections of haptic and visual space are sufficiently well behaved, one can derive the corresponding Riemannian metrics. These metrics impose a unique relation between concepts such as distance, straightness and parallelity. As a consequence, one can predict the frontoparallel plane horopters, parallel- and distance-alleys and egocentric distance relations. In this study, we will determine the affine connections and derive the metrics for the haptic and visual modalities. We will restrict ourselves to the case where the deviations are independent of the reference orientation, and thus will

not take the oblique effect into account. Despite this restriction, our models are still relevant, at least for the visual modality, because many visual experiments described in the literature were conducted in environments without external references. Consequently, the predictions derived from our model can be compared directly to the results reported in the literature.

2. Parallelism in affinely connected spaces

The assumption that visual space is a metrical space is valid if and only if some sort of distance relation can be defined. Blank (1958) proved that a sense of length ordering (i.e. which distance interval is longer) and a sense of alignment (i.e. are three separate points collinear) are, in principle, sufficient for the existence of such a distance relation and, thus, a metric. The assumption that visual space is Riemannian is even more restrictive (Stoker, 1969). In a Riemannian space all geometric concepts such as parallelism and straightness follow from its metric. However, these concepts do not require the existence of a metric: the so-called affinely connected spaces are equipped with concepts of parallelism and straightness without presupposing the existence of a metric. Note that every Riemannian space is also an affinely connected space, but that the reverse is not true. The simplest example of a non-Euclidean space that is affinely connected and Riemannian is the surface of a sphere. We will use the spherical geometry as an example for introducing the basic equations of differential geometry (Stoker, 1969).

Suppose a ship is sailing the seas without deviating either left or right. In that case it will trace out a great circle (provided the earth is spherical). So a great circle is the equivalent of a straight line in spherical geometry. However, great circles always intersect each other in two anti-podal points, so straight lines cannot be parallel globally. Clearly, careful definitions of straightness and parallelity are needed in curved spaces. This can be done by making use of the parallel-transport of a vector along a curve. Let $C(s)$ be a curve parametrised by s and $\vec{V}$ a vector at some point of the curve $C(s)$. The parallel-transport of the vector $\vec{V}$ along the curve $C(s)$ is defined by

$$\frac{d\vec{V}}{ds} = 0. \quad (1)$$

So if $\vec{V}_0$ is a vector at the point $C(s_0)$, we obtain a parallel vector $\vec{V}_1$ at $C(s_1)$ by parallel-transport of $\vec{V}_0$ to $C(s_1)$. The tricky part of Eq. (1) is that the coordinate vectors may change when the coordinates are curvilinear or when the space is curved. The change of each coordinate vector is also a vector and can be written as a linear combination of the coordinate vectors. Using the Einstein summation convention, where the sum is taken

over all equal indices, the coefficients Γ_{ij}^k of this linear combination are defined by

$$\frac{\partial \vec{e}_i}{\partial x^j} = \Gamma_{ij}^k \vec{e}_k, \quad (2)$$

where the $\vec{e}_i$ are the coordinate vectors. The coefficients Γ_{ij}^k are said to define the *affine connection* and spaces equipped with an affine connection are said to be *affinely connected*.

If we substitute $\vec{V} = V^k \vec{e}_k$ in Eq. (1) and use Eq. (2), we find, after re-arranging the terms, the following equation for the change of the components of $\vec{V}$:

$$\frac{\partial V^k}{\partial s} = -\Gamma_{ij}^k V^i \frac{\partial x^j}{\partial s}. \quad (3)$$

So the affine connection determines how the coordinate vectors change as a function of position and, consequently, how the components V^k change: they must change by the same amount in the opposite direction in order for $\vec{V}$ to remain parallel.

The equivalent of a straight line in curved spaces is called a geodesic and it is defined as a line which extends itself in the same direction. Mathematically, this means that a geodesic parallel-transport its own tangent vector. Let $C(s) = (x^1(s), \dots, x^n(s))$ be a curve with $\vec{U} = \frac{\partial x^i(s)}{\partial s} \vec{e}_i$ the unit tangent vector, then $C(s)$ is a geodesic when $\vec{U}$ satisfies Eq. (1), yielding

$$\frac{\partial^2 x^k}{\partial s^2} + \Gamma_{ij}^k \frac{\partial x^i}{\partial s} \frac{\partial x^j}{\partial s} = 0. \quad (4)$$

Eq. (1) is valid for any path, but in general it is not true that the parallel-transport of a vector from A to B along different paths will yield the same vector at B . For example, if we parallel-transport a vector pointing north along the equator from 0° to 180° longitude, it is still pointing north. However, if we move along the 0° meridian, the north pole and the 180° meridian, it is pointing south. In fact, the difference between the vector obtained after its parallel-transport along a closed path and the starting vector is a measure of the curvature of the enclosed area. The local curvature is captured by taking the limit of the enclosed area to zero. This yields the Riemann curvature tensor the components of which are given by

$$R_{jkl}^i \equiv \Gamma_{jl,k}^i - \Gamma_{jk,l}^i + \Gamma_{sk}^i \Gamma_{jl}^s - \Gamma_{sl}^i \Gamma_{jk}^s, \quad (5)$$

where a shorthand notation is used for partial derivatives, defined by

$$f_{,i} \equiv \frac{\partial f}{\partial x^i}. \quad (6)$$

Another important property that can be derived from the affine connection is the so-called *torsion* tensor the components of which are given by

$$T_{ij}^k \equiv \Gamma_{ij}^k - \Gamma_{ji}^k. \quad (7)$$

In Euclidean and Riemannian geometries the components of the torsion tensor are zero. This means that if

one would trace out a rectangle by first moving in the x^1 and then in the x^2 direction and back by the same amounts but first in the x^1 and then in the x^2 direction, one ends up at the starting position. In a space with torsion this is generally not the case.

3. Parallelism and the Riemannian metric

Thus far, we have not used the existence of any metric in our equations, so these equations are valid without the existence of a metric. However, if the affine connection is sufficiently well behaved, a Riemannian metric exists: the coefficients Γ_{ij}^k must be continuous differentiable functions of the coordinates and the space must be torsion free ($T_{ij}^k = 0$, Eq. (7)).

The metric of a Riemannian space, with coordinates $(x^1, \dots, x^n)$, is expressed in terms of its local derivatives. The infinitesimal distance ds is given by

$$(ds)^2 = g_{ij} dx^i dx^j, \quad (8)$$

where the coefficients g_{ij} are real single-valued differentiable functions of the coordinates. Furthermore, the equations $g_{ij} = g_{ji}$ and $\det(g_{ij}) > 0$ must hold. The functions g_{ij} completely determine the Riemannian metric and they are the components of the metric tensor $\mathbf{g}$.

In Riemannian geometries the affine connection is uniquely determined by the metric (Stoker, 1969)

$$\Gamma_{ij}^k = \left\{ \begin{matrix} k \\ ij \end{matrix} \right\} \equiv \frac{1}{2} g^{ks} (g_{sj,i} + g_{is,j} - g_{ij,k}), \quad (9)$$

where g^{ks} is the inverse matrix of g_{ks} and is defined by $g^{ks} g_{sl} = \delta_l^k$,

and δ_l^k is Kronecker's delta which equals 1 if $k = l$ and 0 otherwise. The quantities $\left\{ \begin{matrix} k \\ ij \end{matrix} \right\}$ are called *Christoffel symbols*. Due to the symmetry of the metric ($g_{ij} = g_{ji}$) the Christoffel symbols are also symmetric ($\Gamma_{ij}^k = \Gamma_{ji}^k$), in other words, the space is torsion free.

In order to solve the metric g_{ij} from Eq. (9), it is useful to write this equation in another form

$$g_{ij,k} = \Gamma_{ik}^s g_{sj} + \Gamma_{jk}^s g_{is}. \quad (10)$$

4. Determining the affine connection from the parallelity tasks

In the parallelity tasks, as reported in Cuijpers et al. (2000b), Kappers and Koenderink (1999) and Kappers (1999), subjects had to adjust the orientation of a test bar until it appeared (visually or haptically) parallel to a reference bar for various positions of the bars. Both bars were placed in the horizontal plane (visually at eye height and haptically at waist height) and rotated about the vertical axis through their centres. In order to obtain

the affine connection from the parallelity tasks, we need to formalise our results: Firstly, we interpret the physical Cartesian coordinates as unknown curvilinear coordinates $x^1, \dots, x^n$ (with $n = 2$) in visual or haptic space. Secondly, we associate the orientations of the test and reference bars with vectors of unknown length, denoted by $\vec{V}$ and $\vec{V}_{\text{ref}}$, respectively. So, when the bars appear to be parallel and the perceptual space is affinely connected, the vectors $\vec{V}$ and $\vec{V}_{\text{ref}}$ will also be parallel and one may be obtained from the other by parallel-transport along some curve. With Eq. (3) the affine connection can be calculated from the change in orientation as a function of the bar positions. Subsequently, the metric, the curvature and the geodesics can be calculated using Eqs. (10), (5) and (4), respectively.

Let us start by introducing a convenient representation of the experimental results. Suppose that for a given orientation and position of the reference bar ($\vec{V}_{\text{ref}}$) the test bar ($\vec{V}$) appears to be parallel when its orientation is rotated α radians with respect to the reference bar. In general, the angle α will be a function of the coordinates of the test bar $\alpha = f(x^1, \dots, x^n)$, or in short, $\alpha = f(\vec{x})$. We can write

$$\vec{V}(\vec{x}) = A(\vec{x}) \mathbf{R}(f(\vec{x})) \cdot \vec{V}_{\text{ref}}, \quad (11)$$

where $A(\vec{x})$ denotes the length change as a function of the coordinates and $\mathbf{R}(\alpha)$ is the rotation matrix for an angle α , which is defined as

$$\mathbf{R}(\alpha) = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix}. \quad (12)$$

In tensor notation Eq. (11) can be written as

$$V^k(\vec{x}) = A(\vec{x}) R_l^k(f(\vec{x})) V_{\text{ref}}^l, \quad (13)$$

where the coefficients of R_l^k correspond to the k th row and the l th column of the rotation matrix $\mathbf{R}$ (see Eq. (12)). Note that this is an empirical equation with the sole purpose of describing the data in a concise and convenient way.

By definition, the vectors $V^k(\vec{x})$ will be parallel and therefore Eq. (13) must satisfy Eq. (3). We obtain the change of the components V^k along some curve $C(s) = (x^1(s), \dots, x^n(s))$ by taking the partial derivative of Eq. (13) to s :

$$\frac{\partial V^k}{\partial s} = \frac{\partial A(s)}{\partial s} R_l^k(f(s)) V_{\text{ref}}^l + A(s) \frac{\partial R_l^k(f(s))}{\partial s} V_{\text{ref}}^l, \quad (14)$$

where $A(s) = A(\vec{x}(s))$ and $f(s) = f(\vec{x}(s))$. The equation for parallel-transport is also given in terms of the affine connection in Eq. (3) and by using Eq. (13) it can be expressed in terms of the reference vector V_{ref}^l . Substituting this in Eq. (14) and eliminating V_{ref}^l we get

$$\frac{1}{A(s)} \frac{\partial A(s)}{\partial s} R_l^k(f(s)) + \frac{\partial R_l^k(f(s))}{\partial s} = -\Gamma_{ij}^k \frac{\partial x^j}{\partial s} R_l^i(f(s)). \quad (15)$$

This equation for the affine connection can be written in explicit form by using the chain rule

$$\frac{\partial}{\partial s} = \frac{\partial x^j}{\partial s} \frac{\partial}{\partial x^j},$$

and the fact that the rotation matrix $\mathbf{R}(\alpha)$ has an inverse $\mathbf{R}^{-1}(\alpha) = \mathbf{R}(-\alpha)$ such that

$$R_i^k(\alpha) R_m^l(-\alpha) = \delta_m^k.$$

We obtain

$$\Gamma_{ij}^k = -\frac{\partial \ln A(\vec{x})}{\partial x^j} \delta_i^k - \frac{\partial R_i^k(f(\vec{x}))}{\partial x^j} R_i^l(-f(\vec{x})). \quad (16)$$

In Cartesian coordinates the rotation matrix $R_i^k(\alpha)$ depends only on α and not on the coordinates. In that case, we can write

$$\Gamma_{ij}^k = -\frac{\partial \ln A(\vec{x})}{\partial x^j} \delta_i^k - \frac{\partial f(\vec{x})}{\partial x^j} \varepsilon_i^k \quad (17)$$

with

$$\varepsilon_i^k = \frac{\partial R_i^k(f)}{\partial f} R_i^l(-f). \quad (18)$$

In two dimensions, the rotation matrix $R_i^k(\alpha)$ is given by Eq. (12). In that case the quantities ε_i^k take a very simple form, i.e. $\varepsilon_1^1 = -\varepsilon_2^2 = 1$ and $\varepsilon_1^2 = \varepsilon_2^1 = 0$. So we have for the affine connection (Eq. (17)):

$$\begin{aligned} \Gamma_{1j}^1 &= \Gamma_{2j}^2 = -\frac{\partial \ln A(\vec{x})}{\partial x^j}, \\ \Gamma_{1j}^2 &= -\Gamma_{2j}^1 = -\frac{\partial f(\vec{x})}{\partial x^j}. \end{aligned} \quad (19)$$

Since we did not measure the length change $A(\vec{x})$, the affine connection cannot be fully determined. However, by assuming that the space is torsion free ($\Gamma_{ij}^k = \Gamma_{ji}^k$) we can solve $A(\vec{x})$ using Eq. (19) and obtain

$$\begin{cases} \frac{\partial \ln A(\vec{x})}{\partial x} = \frac{\partial f(\vec{x})}{\partial y} \\ \frac{\partial \ln A(\vec{x})}{\partial y} = -\frac{\partial f(\vec{x})}{\partial x} \end{cases} \Rightarrow \begin{cases} \Delta \ln A(\vec{x}) = 0, \\ \Delta f(\vec{x}) = 0, \end{cases} \quad (20)$$

where $x = x^1$ and $y = x^2$ and where Δ is the Laplace operator. So, if the perceptual space is torsion free, $\ln A(\vec{x})$ and $f(\vec{x})$ both must satisfy Laplace's equation. The general solutions for the Eq. (20) can be obtained by standard calculus given appropriate initial values and boundary conditions.

5. Visual modality

5.1. Experimental results

In the visual parallelity task it was found that the deviation between the orientation of the test bar and the reference bar was independent of the radial distance r of both the test and reference bar (Cuijpers et al., 2000b). On the other hand, the deviations depended linearly on the separation angle ($\phi - \phi_{\text{ref}}$) between the bars. An

example of these data is reproduced in Fig. 1. In the left column the raw data are shown graphically for several test bar locations and a single location (1.47 m, -30°) and a 30° orientation of the reference bar. In the right column the deviations for several locations of the reference bar are combined and plotted as a function of the separation angle (top) and radial distance (bottom). The slopes of the regression lines are 0.34 ± 0.03 and $-2.2 \pm 1.7^\circ/\text{m}$, respectively, but the latter is not significantly different from zero ($p = 0.23$). Note that when the separation between the bars is expressed as arc length instead of the separation angle the slope would become $13 \pm 1^\circ/\text{m}$ which is much larger (in absolute sense) than the $-2.2^\circ/\text{m}$ in the radial direction.

The slope σ of the linear dependence on the separation angle was found to be approximately independent of the location and orientation of the reference bar. The mean slope value differed for each subject but was typically about $\sigma \approx 0.3$. When the reference bar was physically parallel to the walls of the room, however, the slopes were significantly reduced or even zero (independent of the location of the reference bar). It turns out that this 'oblique effect' is due to external reference frames which were still visible in the experiment and it will probably disappear without them

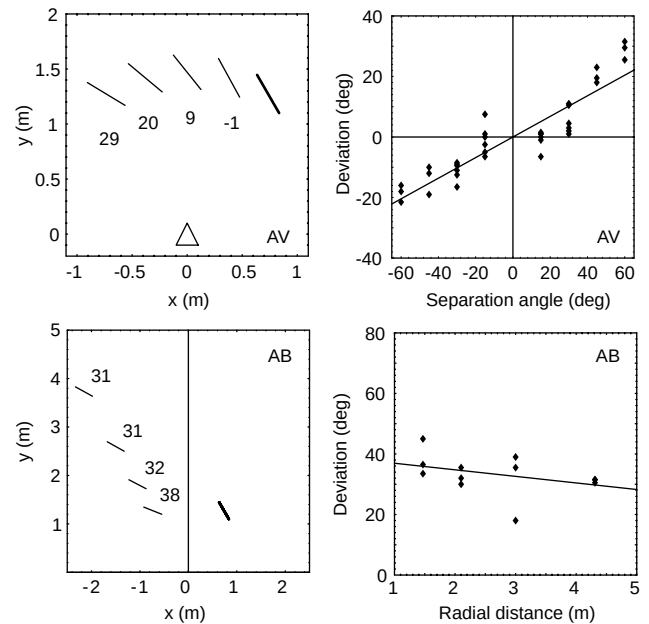


Fig. 1. Example of the data reproduced from Cuijpers et al. (2000b). In each graph of the left column the mean orientation ($n = 3$) of the test bar is indicated for four locations (thin lines) and for a 30° orientation of the reference bar at 1.47 m distance (thick line). The deviations from parallel are indicated in degrees next to each line. In the right column the deviations are plotted as a function of the separation angle between the bars (top) and of the radial distance of the test bar (bottom). The solid lines are linear fits to the data with slopes 0.34 ± 0.03 ($R^2 = 0.83$) and $-2.2 \pm 1.7^\circ/\text{m}$ ($R^2 = 0.14$), respectively.

(Cuijpers et al., 2001). In that case the slope σ will be a subject dependent constant of about $\sigma \approx 0.3$. We can summarise these experimental results in the following formula:

$$f(\phi) = \sigma(\phi - \phi_{\text{ref}}), \quad (21)$$

where $f(\phi)$ corresponds to the deviation from parallel as a function of the test bar location and which is independent of the radial distance r . For the corresponding length change $A(r, \phi)$, no experimental data are available because the bars were of fixed size. However, by assuming that visual space is torsion-free ($\Gamma_{ij}^k = \Gamma_{ji}^k$) we can solve the length change from Eqs. (20) and (21) yielding

$$A(r) = c \left(\frac{r}{r_0} \right)^\sigma, \quad (22)$$

where c is an arbitrary constant and r_0 is the reference distance. Note that the length change is independent of the ϕ -coordinate. The constant c can be solved by requiring that the geometry is Euclidean when the bars are set physically parallel. In other words, in the limit of $\sigma \rightarrow 0$, the length change $A(r)$ should vanish identical to 1, which implies that $c = 1$. By using the chain rule and $x^1 = r \cos \phi$ and $x^2 = r \sin \phi$, the Christoffel symbols are given by (see Eq. (19)):

$$\begin{aligned} \Gamma_{11}^1 &= \Gamma_{12}^2 = \Gamma_{21}^2 = -\Gamma_{22}^1 = -\frac{\sigma}{r} \cos(\phi), \\ \Gamma_{22}^2 &= \Gamma_{12}^1 = \Gamma_{21}^1 = -\Gamma_{11}^2 = -\frac{\sigma}{r} \sin(\phi). \end{aligned} \quad (23)$$

5.2. Metric, curvature and geodesics

Once the Christoffel symbols are determined, the metric can be obtained from Eq. (10). Assuming that in the limit of $\sigma \rightarrow 0$ the metric is Euclidean, we obtain

$$g_{ij} = \left(\frac{r}{r_0} \right)^{-2\sigma} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (24)$$

where r_0 is an arbitrary distance for which the metric is Euclidean. The metric in Eq. (24) is conformal, that is, the angles between vectors defined by this metric are equal to the angles defined by the Euclidean metric.

The curvature tensor follows from Eq. (5) and by substituting Eq. (23) we find $R_{jkl}^i = 0$ for all i, j, k , and l . The zero curvature expresses the experimental observation that when bars are subsequently set parallel to each other along a closed loop, no difference in orientation is found after the loop is being circumnavigated. A zero curvature corresponds to a flat Euclidean space and, as a result, coordinates ξ and η can be found which are Cartesian in visual space and curvilinear in physical space. For these coordinates the metric is equal to the identity matrix, so a coordinate transformation is needed which transforms Eq. (24) into the identity matrix. It can be shown that the following coordinates

have this property:

$$\begin{aligned} \xi &= \frac{r_0}{1-\sigma} \left(\frac{r}{r_0} \right)^{1-\sigma} \cos((1-\sigma)\phi), \\ \eta &= \frac{r_0}{1-\sigma} \left(\frac{r}{r_0} \right)^{1-\sigma} \sin((1-\sigma)\phi), \end{aligned} \quad (25)$$

where r and ϕ are the polar coordinates defined in the usual way. By virtue of the fact that ξ and η are Cartesian coordinates in visual space, lines of constant ξ and lines of constant η are straight, parallel, equidistant lines.

The geodesics are found by solving the set of differential equations given in Eq. (4). Finding the general solution in Cartesian or even polar coordinates is rather tedious. Fortunately, the solutions of Eq. (4) are trivial in ξ, η -coordinates. In these coordinates the metric reduces to the identity matrix and, consequently, all Christoffel symbols are zero (this follows from Eq. (10)). In that case, Eq. (4) reduces to the statement that the second order derivatives of ξ and η are zero, and the general solution is the set of straight lines in the ξ - η plane:

$$a\xi + b\eta = 1, \quad (26)$$

with a and b arbitrary constants.

5.3. Comparison with the literature

5.3.1. Egocentric distance

The egocentric distance can be predicted from the parallelity task by making use of Eq. (24). The perceived distance $d(r)$ is

$$d(r) = \int_0^r \sqrt{g_{ij} \frac{\partial x^i}{\partial s} \frac{\partial x^j}{\partial s}} ds = \frac{r_0}{1-\sigma} \left(\frac{r}{r_0} \right)^{1-\sigma}. \quad (27)$$

In Fig. 2 the distance function $d(r)$ is plotted for $\sigma = 0.3$ and $r_0 = 1$ m (solid line) with r ranging from 0 to 10 m. The constant r_0 is chosen arbitrarily. Several distance functions have been reported in the literature that were obtained experimentally. For example, Gilinsky (1951) measured the successive distance intervals that appeared to be equal to a “subjective memorized meter stick” in an indoor archery range of about 25 m long. The long-dashed line in Fig. 2 shows her results for observer JB and corresponds to $28.5r/(28.5 + r)$. She found similar results in a bisection task conducted on a lawn outdoors with distances ranging up to 60 m. In a study by Wagner (1985) observers had to judge the distance between two stakes in a large open field. The distance of the stakes from the observer ranged from 7 to 72 m. The resulting distance function (from the pooled data of 5 observers) is indicated by the short-dashed line with equation $1.305(0.49r)^{0.97}$. The dash-dotted line corresponds to Luneburg’s function $2/\sqrt{-K} \operatorname{arctanh}(\sqrt{-K}e^{-\sigma l/r})$ with $K = -0.98 \text{ m}^{-2}$ the curvature of visual space, $\sigma = 9.7$ a

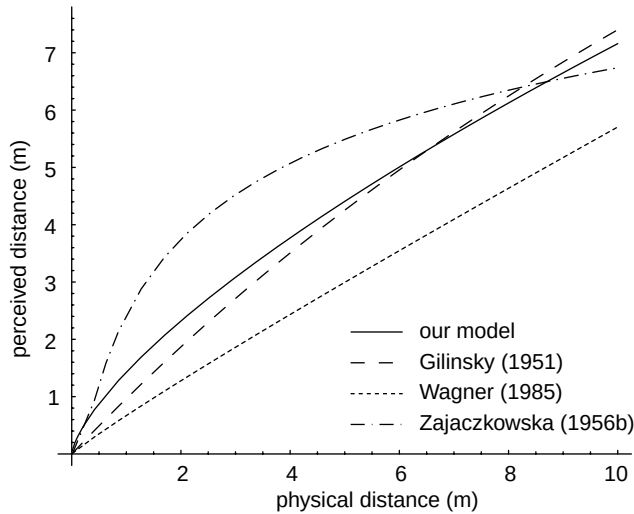


Fig. 2. Perceived egocentric distance as a function of the physical distance. The solid line corresponds to the prediction of our model for $r_0 = 1$. The dashed lines indicate results from the literature.

personal constant and $I = 0.063$ m the interpupillary distance. These values were obtained experimentally by Zajaczowska (1956a) for subject HS. In this study, the task was to match distances between pairs of dim lights in complete darkness at distances up to 3 m from the subject. Luneburg's function is expressed in arbitrary units and, therefore, is determined up to an arbitrary constant.

As can be seen from Fig. 2, the function $d(r)$ resembles Gilinsky's function very well (long dash). Both functions are always decelerating, but our distance function goes to infinity as r goes to infinity whereas Gilinsky's function approaches the value 28.5 asymptotically. This difference plays a role only for very large distances, however. The agreement between Gilinsky's function and our function (Eq. (27)) is mainly dependent on the choice of the constant r_0 . Choosing a value of 0.3 m would result in a function that is in reasonable agreement with Wagner's function (short dash). However, it is not possible to achieve a reasonable agreement with Luneburg's function (dash-dot) because it first increases steeply and then levels off to a constant value. Thus, by a suitable choice of r_0 , Eq. (27) can be brought into good agreement with either Wagner's or Gilinsky's function but not with Luneburg's function, at least for the range of 1.47–4.31 m that was used in the parallelity task (see Cuijpers et al., 2000b).

5.3.2. Frontoparallel horopters

The frontoparallel horopters are given by lines of constant ξ in Eq. (25) and correspond to lines which are concave towards the observer. This is shown in Fig. 3 where the observer is located at the origin and where the x -axis coincides with the straight-ahead direction. The curvature κ is clearly not constant, but may be

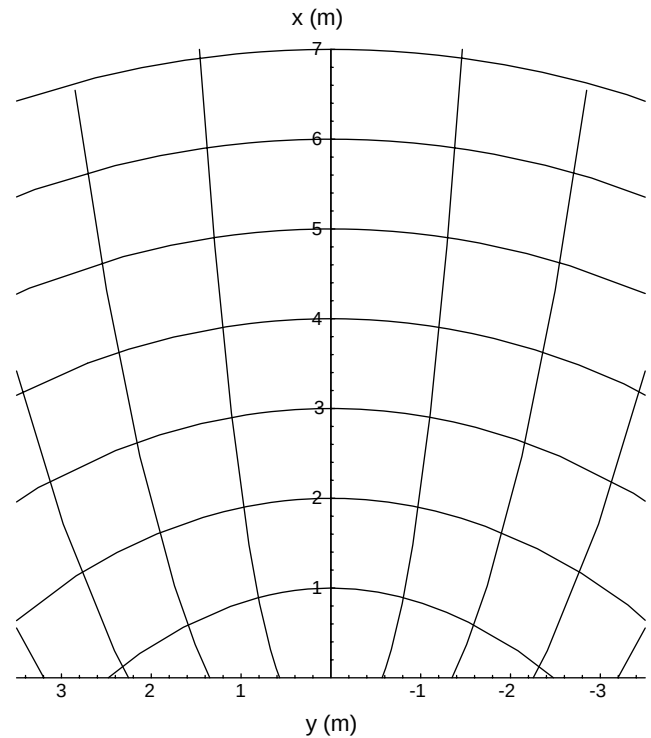


Fig. 3. Frontoparallel horopters and parallel-alleys for the visually perceived space.

calculated for the centre ($y = 0$), yielding

$$\kappa = -\frac{1 - \sigma}{r}, \quad (28)$$

where the negative sign is defined as concave towards the observer.

From Eq. (28) it is clear that the curvature κ decreases with the distance, but it will never become positive. The curvatures reported in the literature vary tremendously among the different studies. For example, for a distance of 1.5 m the observed curvatures ranged from -0.9 to 3 m^{-1} (Shipley, 1959). Often the frontoparallel horopters are skew-symmetric about the median line (Battro et al., 1976; Ogle, 1964) unless the design of the experiment was symmetric (Zajaczowska, 1956b). Several authors have demonstrated inversions from concave horopters to convex horopters (e.g. Zajaczowska, 1956b) or multiple inversions (e.g. Battro et al., 1976), which contradict Eq. (28). However, the distance at which the inversion occurs differs considerably across subjects and tasks. Moreover, in most other studies the inversion point is not observed (e.g. Helmholtz, 1962; Ogle, 1964; Shipley, 1959). In most studies the curvature is an approximately monotonically increasing and decelerating function of the distance. The same is true for Eq. (28). The circumstances under which the experimental results were obtained also varied tremendously: For example, Helmholtz (1962) used three fine, black silk threads suspended from pegs in a horizontal

beam. The outer two threads were fixed symmetrically about the median plane at 15–75 cm from the observer. The third, central thread was in the median plane, several millimeters behind the plane of the other two threads. Zajaczkowska (1956b) replaced the threads by dim lights at eye-level in a dark room with distances ranging from 0.5 to 3 m. The distance of the central light was adjusted. Ogle (1964) used rods of constant visual angle in dim room lighting conditions at distances between 20 and 75 cm and at 6 m. In this study the central rod was fixed and 4 rods on each side of the median plane were adjusted along lines of sight. Battro et al. (1976) used orange stakes of constant visual angle in a large open field in full-cue conditions at distances

ranging from 0.5 to 45 m. Both Zajaczkowska's and Ogle's task were employed.

5.3.3. Parallel- and distance-alleys

Classically, the parallel-alleys were constructed by arranging pairs of dim lights in the dark in a horizontal plane at eye height until they appeared to form two parallel and straight lines symmetric about the median plane (Blumenfeld, 1913; Hardy, Rand, Rittler, Blank, & Boeder, 1953; Indow, Inoue, & Matsushima, 1962; Zajaczkowska, 1956b). The most distant pair of stimuli was always fixed and the other pairs were added successively at increasingly closer distances from the observer. The separation between the nearer pair was

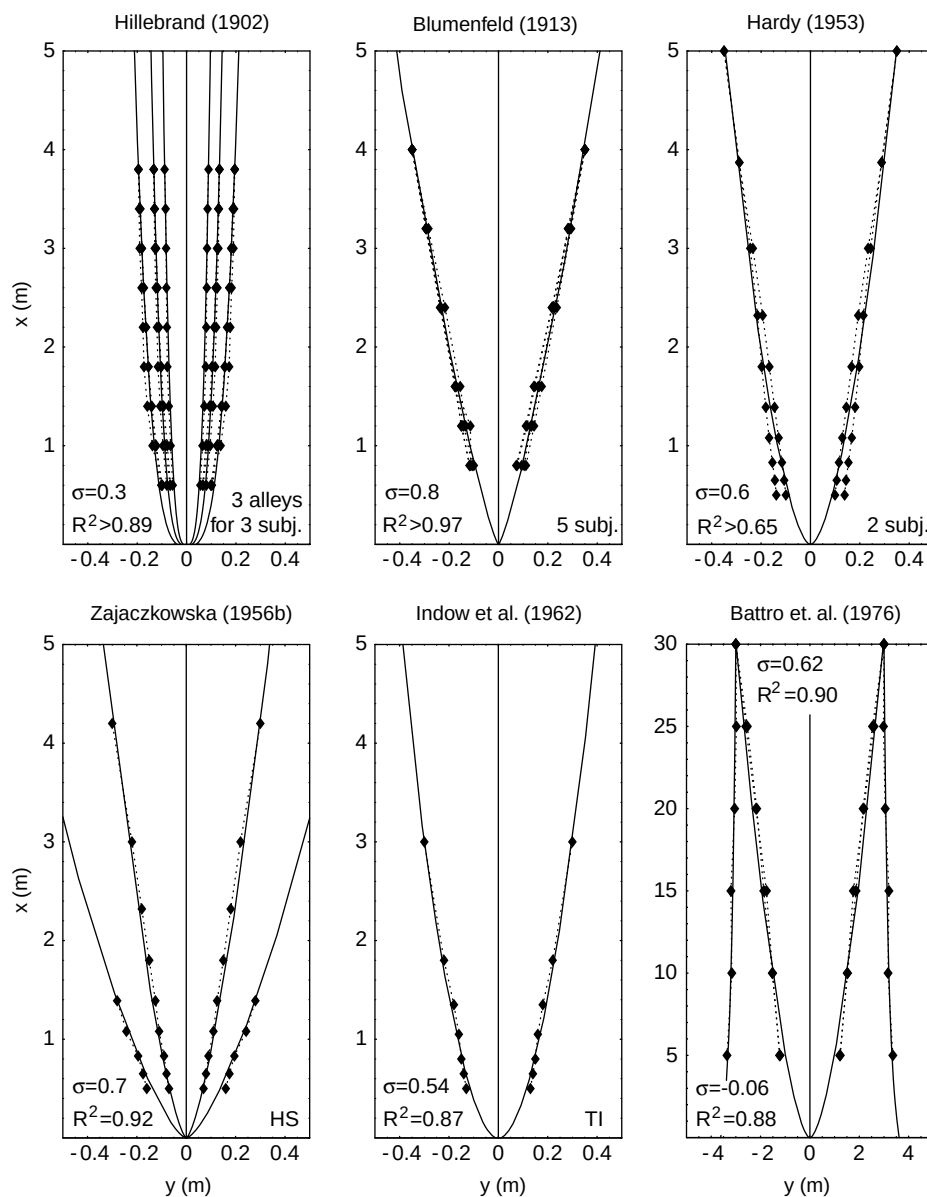


Fig. 4. Experimental data from the literature for the parallel-alleys. Each graph is a top view of the settings with the observer at the origin. The data points indicate the stimulus positions and the solid lines show the fits using our model. The fitting parameters σ and the squared correlation coefficients are indicated in each graph.

adjusted while all previously adjusted pairs remained visible. Subjects were often allowed to readjust their settings after completing the alley. Typically, the heads of the subjects were fixed and free eye movements were allowed. Examples of the obtained parallel-alleys are shown in Fig. 4. Instead of lights in the dark, Hillebrand (1902) used black threads against a white background in dim room lighting conditions and for various fixation strategies. Entirely different circumstances were present in a study by Battro et al. (1976) who conducted his experiments out of doors on large polo fields using yellow rods of fixed height (1 or 1.5 m). In this study the experimenter and the ground plane were visible, and subjects were allowed to move their heads. The distance-alleys were constructed in a similar fashion but now the pairs of threads/lights/rods were arranged so that they appeared to be separated by the same distance. However, only two pairs were visible at the same time, i.e. a distant fixed pair (the same as for the parallel-alley) and a nearer adjustable pair. Usually, the parallel-alleys are lying inside of the distance-alleys.

The lines of constant η in Eq. (25) correspond to lines which are parallel to the median line. The parallel-alleys are formed by the sets $\eta = \pm c$ (where c is an arbitrary constant) and converge a little towards the observer (see Fig. 3). If we require these lines to pass through the most distant fixed pair of stimuli, we can fit our equations to the data from the parallel-alleys with only one parameter, i.e. the slope σ . Let the coordinates of the most distant pair be $(x_0, \pm y_0)$, then the equation for the parallel-alley becomes

$$r^{1-\sigma} \sin((1-\sigma)\phi) = r_0^{1-\sigma} \sin((1-\sigma)\phi_0), \quad (29)$$

where $r_0 = \sqrt{x_0^2 + y_0^2}$, $\phi_0 = \arctan(\frac{y_0}{x_0})$ and $-\frac{\pi}{2} \leq \phi \leq \frac{\pi}{2}$. In the parallel-alley experiments the x -coordinate (corresponding to the distance from the observer) was typically fixed and the y -coordinate was being adjusted. Therefore we fitted Eq. (29) by minimising the y -error as a function of σ . The results for various studies are shown in Fig. 4. As can be seen, the fits are quite good with squared correlation coefficients of $R^2 = 0.65$ and higher (see Fig. 4). The fitted parameter σ is different for each subject but for a given subject it is very similar for different sizes of the parallel-alleys: $\sigma = 0.6$ for the three alleys for all three subjects in Hillebrand's study; $\sigma = 0.7$ for subject HS. in Zajackowska's study. In Battro's study both converging and diverging alleys were found resulting in negative values for the slope, i.e. $\sigma = -0.06$. In the parallelity task the values for σ ranged from 0.15 to 0.6 depending on the subject, so the order of magnitude is similar.

Since the intrinsic curvature is zero in our model, the equidistance-alleys are equal to the parallel-alleys, which contradicts the reports in the literature (e.g. Battro et al., 1976; Blumenfeld, 1913; Hardy, Rand, & Rittler, 1951; Indow et al., 1962; Indow, 1991, 1997; Zajackowska,

1956a,b). However, the differences between the equidistance- and the parallel-alleys are very small and almost negligible beyond a range of 1.5 m, which is the range used for the parallelity task. Moreover, the difference between parallel- and distance-alleys is probably an artifact due to the different methods for constructing them (Ehrenstein, 1977).

5.3.4. Equidistance circles

The egocentric distance $d(r)$ given in Eq. (27) is independent of the direction. As a result, points which appear equidistant from the observer lie on circles centred on the observer. Luneburg (1947) assumed that points lying on a Vieth–Müller circle would look equidistant. A direct test by Foley (1966) revealed that points of apparent equidistance actually lie in between the egocentric circles and Vieth–Müller circles.

6. Haptic modality

6.1. Experimental results

In Kappers and Koenderink (1999) and Kappers (1999) it was found that the orientation of the test bar for which it appears haptically parallel to the reference bar deviated systematically from physically parallel: the deviation from parallel increased linearly with the lateral distance (y direction) between the bars at a rate of about $\sigma = 30^\circ/\text{m}$ but was independent of the distance in depth (x direction). Typical results are shown in Fig. 5. The results of the collinearity task (Kappers & Koenderink, 1999) are consistent with the parallelity task. Consequently, these results can be described by the same model.

The results from the haptic version of the parallelity task can be summarised in the following equation:

$$f(x, y) = \sigma(y - y_{\text{ref}}), \quad (30)$$

where $f(x, y)$ is the change in orientation of the test bar as a function of its position and depends only on the lateral distance y . As in the visual modality, the corresponding length change $A(x, y)$ is unknown because the size of the bars was fixed in the experiments. By assuming that the haptic space is torsion-free ($\Gamma_{ij}^k = \Gamma_{ji}^k$) the length change $A(x, y)$ can be determined from the gradient $f(x, y)$ by using Eq. (20). The length change becomes

$$A(x, y) = \exp(c + \sigma(x - x_0)), \quad (31)$$

where c is an arbitrary constant and x_0 the depth of the reference bar. By requiring that the length does not change when there are no deviations ($\sigma = 0$) it follows that $c = 0$. Expressed in the coordinates $x^1 = x$ and

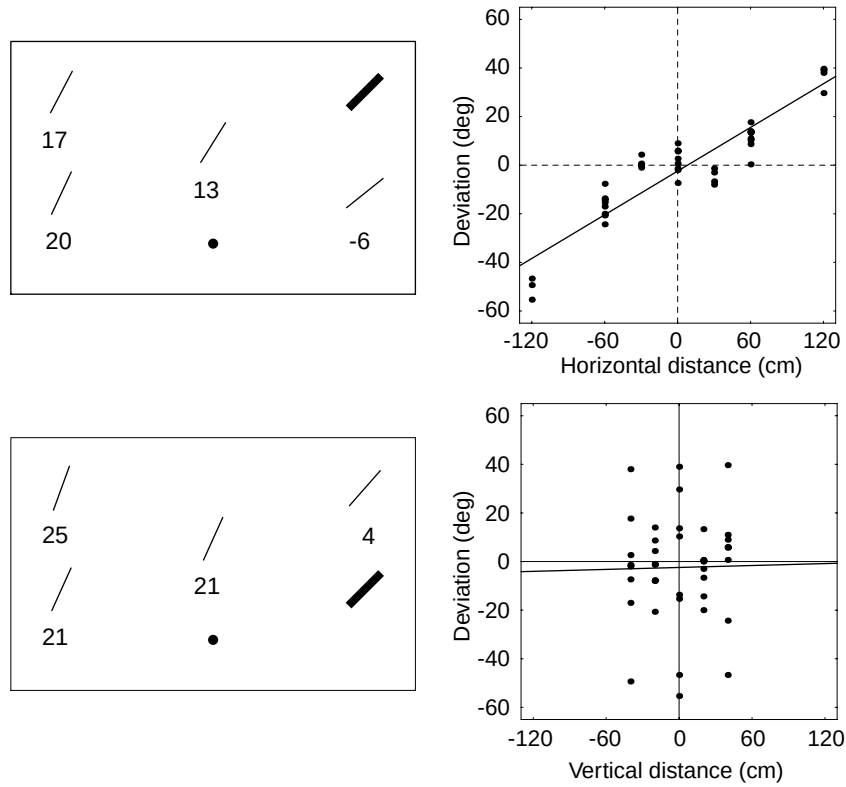


Fig. 5. Example of the data for subject ML reproduced from Kappers (1999). In the left column a graphical and numerical representation of the results are shown for two locations of the reference bar (thick lines). In each graph the mean orientation ($n = 3$) of the test bar is indicated for four locations (thin lines) and for a 45° orientation of the reference bar. The deviations from parallel are indicated in degrees next to each line. In the right column the deviations are plotted as a function of the horizontal distance (top) and of the vertical distance (bottom). The solid lines are linear fits to the data with slopes $30^\circ/\text{m}$ (top) and $-1^\circ/\text{m}$ (bottom).

$x^2 = y$, the Christoffel symbols are:

$$\begin{aligned} \Gamma_{11}^1 &= \Gamma_{12}^2 = \Gamma_{21}^2 = -\Gamma_{22}^1 = -\sigma, \\ \Gamma_{22}^2 &= \Gamma_{12}^1 = \Gamma_{21}^1 = \Gamma_{11}^2 = 0. \end{aligned} \quad (32)$$

6.2. Metric, curvature and geodesics

From Eqs. (10) and (32) the metric can be determined, yielding

$$g_{ij} = \exp(-2\sigma(x - x_0)) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (33)$$

where x_0 is an arbitrary constant corresponding to the depth at which the haptic metric reduces to the Euclidean metric. Just as for the visual modality, the haptic metric is conformal.

Substituting Eq. (32) in Eq. (5), we find the components of the curvature tensor $R_{jkl}^i = 0$. Hence, the intrinsic structure of haptic space is flat, and coordinates u, v exist which are Cartesian in haptic space and curvilinear in physical space. These coordinates can be found most easily by using the fact that the coefficients of the metric in u, v coordinates must be equal to the

identity matrix. Hence, we obtain

$$\begin{aligned} u &= \frac{1}{\sigma} \exp(-\sigma(x - x_0)) \cos(\sigma(y - y_0)), \\ v &= \frac{1}{\sigma} \exp(-\sigma(x - x_0)) \sin(\sigma(y - y_0)), \end{aligned} \quad (34)$$

with y_0 an arbitrary constant corresponding to the symmetry axis $y = y_0$. It seems plausible to take the median line as a symmetry axis, in which case $y_0 = 0$.

In u, v coordinates the metric reduces to the identity matrix, which immediately implies that all the Christoffel symbols are zero in these coordinates (see Eq. (10)). As a result, the solution of Eq. (4) is trivial and the general equation for haptic geodesics is

$$au + bv = 1, \quad (35)$$

where a and b are arbitrary constants.

The lines of constant u correspond to haptic frontoparallel lines and the lines of constant $\pm v$ to haptic parallel-alleys. These lines are shown in Fig. 6 for $x_0 = y_0 = 0$ and $\sigma = 30^\circ/\text{m}$. In the inset the range is restricted to $120 \text{ cm} \times 60 \text{ cm}$, corresponding to the experimental range reported in Kappers and Koenderink (1999) and Kappers (1999). The subject is

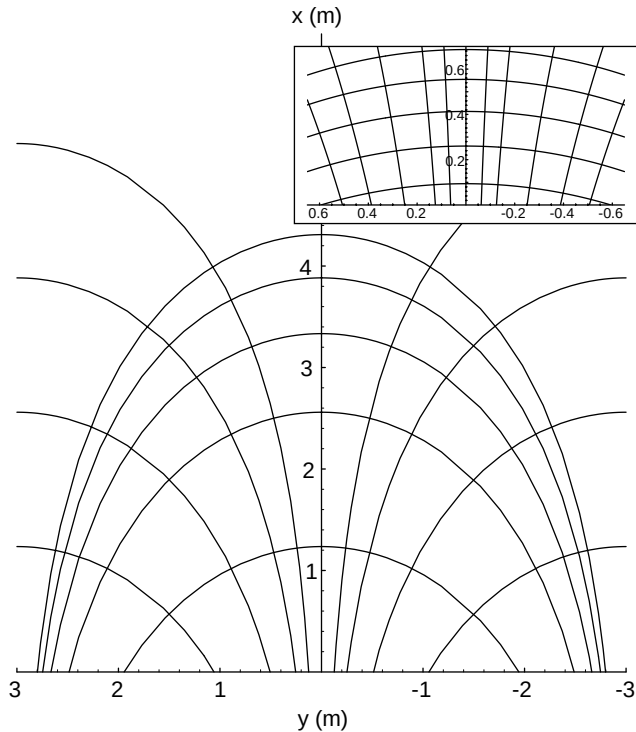


Fig. 6. Frontoparallel horopters and parallel-alleys for the haptically perceived space. In the inset the horopters and alleys are shown for realistic ranges.

located at the origin and the x -axis is the direction straight ahead.

It can be seen in Fig. 6 that the frontoparallel lines are always concave towards the observer. The curvature at the x -axis is constant: by calculating the curvature of lines of constant u for $y=0$ we find $\kappa = -\sigma$. The parallel-alleys approach the x -axis asymptotically. Due to the periodic nature of the equations in Eq. (34) the pattern of Fig. 6 repeats itself in the y direction every $\pi/\sigma k$ with k any whole number. Furthermore, the parallel-alleys are shifted versions of the frontoparallel lines with a shift in y of $\pi/2\sigma$. Unfortunately, this is not liable to experimental testing because it happens beyond the reach of human subjects, as is shown in the inset.

6.3. Egocentric distance and equidistance circles

Analogous to Eq. (27) the egocentric distance can be calculated from the metric for the haptic space, yielding

$$d(r, \phi) = \frac{\exp(\sigma x_0)}{\sigma \cos \phi} (1 - \exp(-\sigma r \cos \phi)), \quad (36)$$

with r and ϕ the polar coordinates. Note that the egocentric distance is direction dependent. Eq. (36) is shown in Fig. 7 for $x_0 = 0$ and for $\phi = 0, \pi/6, \pi/3$, and $\pi/2$. It is clear that the haptically perceived distance is compressed compared to the physical distance and eventually saturates to a constant value of

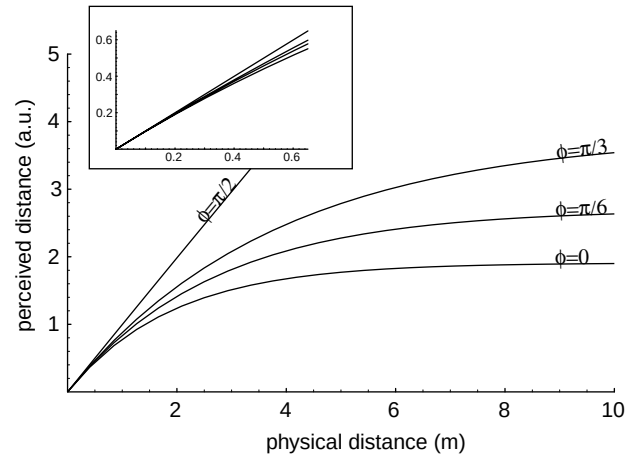


Fig. 7. Haptically perceived distance as a function of the physical distance. In the inset the range is limited to realistic values.

$\exp(\sigma x_0)/(\sigma \cos \phi)$. In the limit of $\phi \rightarrow \pi/2$, Eq. (36) simplifies to $d(r, \pi/2) = r \exp(\sigma x_0)$, so the haptically perceived distance in the lateral direction is not compressed and does not saturate to a constant value.

Eq. (36) predicts that points which appear equidistant from the subject do not lie on circles centered on the subject but lie on curves with an egg-like shape (see Fig. 8). Thus, haptic space is anisotropic, but for realistic ranges the anisotropy is only small (see inset of Fig. 8). However, many subjects have much larger values for the gradient σ up to $75^\circ/\text{m}$. For these subjects the anisotropy could be observable experimentally.

6.4. Comparison with the literature

Compared to the number of studies in vision research only few studies are concerned with haptic perception, let alone haptic space perception. A notable exception is the study of Blumenfeld (1937) who extended his research on visual space to the haptic plane: two needles were fixed on a board symmetrically to the median plane with a 20–25 cm thread attached to each. The task was to grasp each thread with one hand and to straighten them out so that they appeared parallel to each other and to the median line. This could be done either by moving the fingers along each thread or by holding the end of each thread. The haptic parallel-alleys reported by Blumenfeld tend to diverge towards the subject when the separation is smaller than shoulder width and for larger separations the alleys gradually become parallel and slightly convergent (see Fig. 9). This contradicts our model where the alleys typically converge towards the subject.

Blumenfeld also measured the haptic frontoparallel horopters using two setups: in the first setup three rods were used two of which were placed in a frontoparallel plane symmetrically to the median plane and one in between which could be displaced in depth. The other

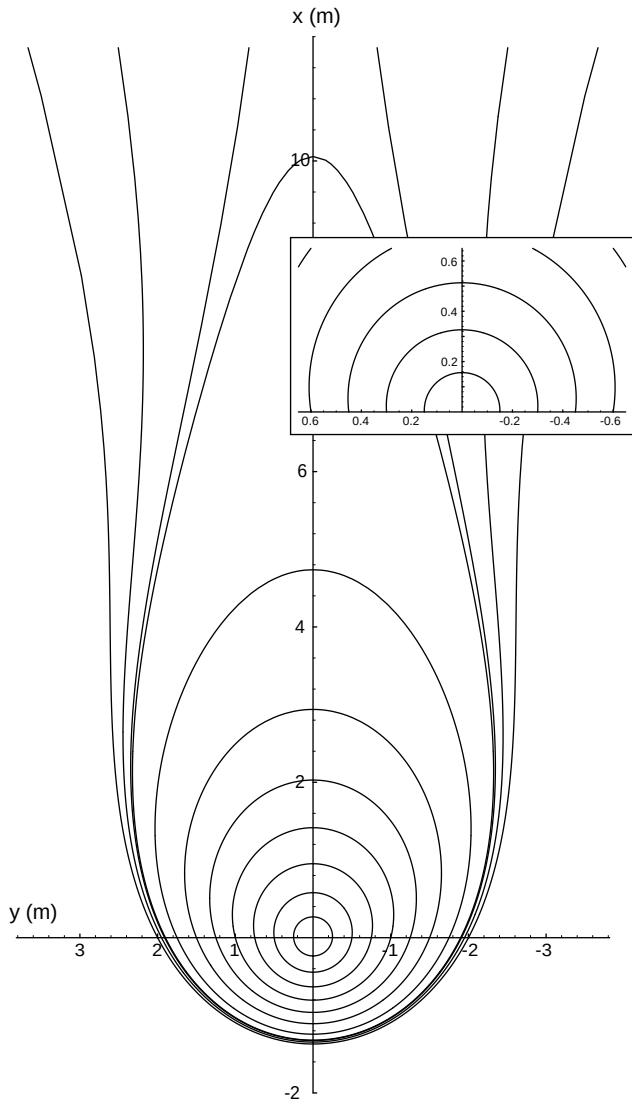


Fig. 8. Equidistance circles in haptic space. The subject is located at the origin, the x -axis being the direction straight ahead. In the inset the equidistance ranges are shown for realistic ranges.

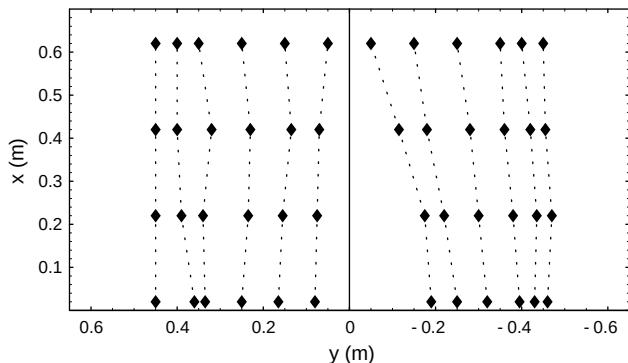


Fig. 9. Example of the haptic parallel-alleys as determined by Blumenfeld (1937) for subject Te.

setup consisted of a steel sheet that could be bent concave or convex towards the subject. He found that the haptic horopters were always concave towards the subject and with a very small variability as a function of the distance. The curvature of the horopters was approximately $\kappa \approx -0.5 \text{ m}^{-1}$ for the setup with the three rods and -0.3 m^{-1} for the setup with the steel band (see Blumenfeld, 1937). These values are remarkably similar to the estimated value $\kappa = -\sigma = -30^\circ/\text{m} \approx -0.52 \text{ m}^{-1}$.

7. Visual versus haptic metric

It seems logical that if an internal representation of the world exists, haptically and visually perceived structures should eventually lead to the same representation. Clearly, this is not the case: for example, lines which appear haptically parallel do not appear visually parallel, etc. (cf. Figs. 3 and 6). On the other hand, the intrinsic structures of both visual and haptic space are flat. This leads to the possibility that the visual space and haptic space can be matched without stretching or bending. In that case a single (internal) representation is possible but with the distinction that the corresponding physical structures depend on the type of perception. Visual and haptic space could be identified as follows:

$$\begin{aligned} u &= a\xi - b\eta + c, \\ v &= b\xi + a\eta + d, \end{aligned} \quad (37)$$

with a, b, c and d arbitrary constants corresponding to an arbitrary scaling, rotation and translation. Substituting Eqs. (34) and (25) yields a mapping from physical to physical space such that the visual appearance of one structure is mapped to another structure with the same haptic appearance.

8. Conclusions

The affine connections of visual space and haptic space have been determined from the data of the parallelity experiments. Hereby, it was assumed that the visually and the haptically perceived spaces are torsion-free. Subsequently, the Riemannian metrics have been derived from the affine connection. Neither model incorporates the experimental observation that the measured deviations are often significantly smaller for non-oblique orientations of the reference bar. In the case of the visual experiments, this oblique effect was due to external references. Thus, the current model for visual space is restricted to environments without external references. This might also be true for the haptic model. Apart from the cases where the deviations are significantly smaller due to the oblique effect, both

models provide a good description of the experimental data.

Luneburg (1947) proposed that visual space is a Riemannian space of constant negative curvature and based his model on the observation that the parallel-alleys lie inside the distance-alleys. On the basis of the parallelity task we find a zero curvature for visual space and, as a consequence, the parallel-alleys and distance-alleys should coincide. However, the measured differences between parallel-alleys and distance-alleys are often very small for distances larger than 1.5 m (see Battro et al., 1976; Hardy et al., 1951; Indow et al., 1962; Indow & Watanabe, 1984; Zajackowska, 1956a). Moreover, the differences between the parallel- and distance-alleys could have arisen from the different methods in constructing them (Ehrenstein, 1977). The various examples of the parallel-alleys reported in the literature can be fitted well by our equations with only one free parameter. The fact that our model is based directly on experimental data is a distinct improvement on Luneburg's model.

The visually perceived egocentric distance in our model is isotropic and always decelerates at a constant rate. Our model is in reasonable agreement with the experimental data obtained outdoors by Gilinsky (1951) and Wagner (1985). However, in previous studies for a pointing task (Cuijpers, Kappers, & Koenderink, 2000a) and a collinearity task (Cuijpers, Kappers, & Koenderink, 2002), we found that the egocentric distance function should have an inflection point: namely, it should have a transition from an accelerating part in near space to a decelerating part in far space. This is in conflict with the present model. Luneburg's distance function does have an inflection point, but nonetheless it is inconsistent quantitatively. The actual locations of apparently equidistant points were measured by Foley (1966) and lie in between the equidistance circles predicted by our model and by Luneburg's model. Luneburg's model was modified by Heller (1997) to incorporate Foley's results.

The visual frontoparallel horopters predicted by our model are always concave towards the observer and their curvatures decrease with distance. The inversion from concave to convex horopters, as reported in the literature, is not predicted. However, the variability between the shapes of the horopters reported in the literature is considerable and in many cases an inversion from concave to convex is not observed (see Shipley (1959) for an overview).

Reports on the structure of haptic space are very sparse. An exception is a study by Blumenfeld (1937) who measured haptic parallel-alleys and frontoparallel horopters. The predicted horopters from our model are in surprisingly good agreement with the findings of Blumenfeld. Unfortunately, the predicted parallel-alleys are not in good agreement but the reason is unclear. On

the other hand, Blumenfeld's conclusions about the haptic alleys were not very consistent with his own data. The structure of haptic space was also measured using a pointing task and a collinearity task (Kappers & Koenderink, 1999). The results were consistent with the parallelity task (Kappers & Koenderink, 1999; Kappers, 1999) and, therefore, consistent with our haptic model.

Both haptic and visual space are distorted in relation to physical space but in a different way. So a given physical structure will have a different appearance when perceived haptically or visually. On the other hand, the intrinsic geometries of both visual and haptic space are Euclidean and both spaces can be matched without bending or stretching. Although our results apply only to the horizontal plane, it was found earlier that visual space is also Euclidean in the frontoparallel plane (Indow & Watanabe, 1988; Indow, 1991). If visual and haptic space are Euclidean, a mapping exists relating the visual shape of an object to an equivalent haptic shape. The corresponding physical shapes will be different but with the same appearance.

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